

Name: _____

1. State Lloyd's algorithm.

[6 points]

State Lloyd's algorithm for finding a clustering of examples $x_1, \dots, x_n \in \mathbb{R}^p$ into k clusters, given initial cluster centers $\mu_1, \dots, \mu_k \in \mathbb{R}^p$. **For full credit:** Briefly describe the algorithm using both words and clear notation.

Solution.

The algorithm. Repeat the following two steps until the assignment stops changing.

1. *Assignment step.* Holding the centers fixed, assign each example to the nearest center: for each $i \in \{1, \dots, n\}$, set

$$c(i) = \arg \min_{j \in \{1, \dots, k\}} \|x_i - \mu_j\|^2,$$

breaking ties by a fixed rule. This defines the groups $C_j = \{i : c(i) = j\}$.

2. *Update step.* Holding the assignment fixed, move each center to the mean of its own group: for each $j \in \{1, \dots, k\}$, set

$$\mu_j = \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

2. Why the objective never increases.

[2 points]

No step of Lloyd's algorithm increases the k -means objective J . Why? **For full credit:** A brief but specific response is enough; I'm not looking for a proof.

Solution. The objective is

$$J(c, \mu) = \sum_{i=1}^n \|x_i - \mu_{c(i)}\|^2,$$

a function of the two arguments c and $\mu = (\mu_1, \dots, \mu_k)$. Each step of Lloyd's algorithm minimizes J *exactly* in one argument while holding the other fixed, and the pre-step value of that argument is one of the candidates the minimum is taken over. So neither step can increase J .